

## SAFETY DATA SHEET

Revision Date 30-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** 2,2,2-Trifluoroethyl vinyl ether

**Cat No. :** L20218

**CAS No** 406-90-6  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                        |            |
|------------------------|------------|
| Flammable liquids      | Category 2 |
| Germ Cell Mutagenicity | Category 2 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor  
Suspected of causing genetic defects

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge

**Response**

IF exposed or concerned: Get medical attention/advice

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                        | CAS No   | Weight % |
|----------------------------------|----------|----------|
| Ethene, (2,2,2-trifluoroethoxy)- | 406-90-6 | <=100    |

### 4. First-aid measures

|                                            |                                                                                                                                              |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                       |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                              |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                            |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                 |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                 |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                        |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                                                               |
| <b>Flash Point</b>                      | -30 °C / -22 °F                                                                                                                                                                                                        |
| <b>Method -</b>                         | No information available                                                                                                                                                                                               |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                                                               |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                                                        |
| <b>Upper</b>                            | No data available                                                                                                                                                                                                      |
| <b>Lower</b>                            | No data available                                                                                                                                                                                                      |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                                                               |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                                                               |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

None known.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
0

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
-

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                            |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep refrigerated. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Oxidizing agent.                                                                                                                                                                                                                                                                |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                           | ACGIH TLV | OSHA PEL | NIOSH                                             | Mexico OEL (TWA) |
|-------------------------------------|-----------|----------|---------------------------------------------------|------------------|
| Ethene,<br>(2,2,2-trifluoroethoxy)- |           |          | Ceiling: 2 ppm<br>Ceiling: 10.3 mg/m <sup>3</sup> |                  |

Legend

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

**Personal Protective Equipment**

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                                |
|----------------------------------------|------------------------------------------------|
| Physical State                         | Liquid                                         |
| Appearance                             | No information available                       |
| Odor                                   | Ether                                          |
| Odor Threshold                         | No information available                       |
| pH                                     | No information available                       |
| Melting Point/Range                    | No data available                              |
| Boiling Point/Range                    | 42 - 43 °C / 107.6 - 109.4 °F                  |
| Flash Point                            | -30 °C / -22 °F                                |
| Evaporation Rate                       | No information available                       |
| Flammability (solid,gas)               | Not applicable                                 |
| Flammability or explosive limits       |                                                |
| Upper                                  | No data available                              |
| Lower                                  | No data available                              |
| Vapor Pressure                         | No information available                       |
| Vapor Density                          | No information available                       |
| Specific Gravity                       | 1.135 g/cm <sup>3</sup>                        |
| Solubility                             | No information available                       |
| Partition coefficient; n-octanol/water | No data available                              |
| Autoignition Temperature               | No information available                       |
| Decomposition Temperature              | No information available                       |
| Viscosity                              | No information available                       |
| Molecular Formula                      | C <sub>4</sub> H <sub>5</sub> F <sub>3</sub> O |
| Molecular Weight                       | 126.08                                         |

## 10. Stability and reactivity

**Reactive Hazard** None known, based on information available

**Stability** Stable under normal conditions.

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Oxidizing agent                                                   |
| <b>Hazardous Decomposition Products</b> | None under normal use conditions                                  |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                          |
| <b>Hazardous Reactions</b>              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                           | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------------|----------|------------|------------|------------|------------|------------|
| Ethene,<br>(2,2,2-trifluoroethoxy)- | 406-90-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

## 13. Disposal considerations

**Waste Disposal Methods**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT**

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.  
 Technical Name (2,2,2-Trifluoroethyl vinyl ether)  
 Hazard Class 3  
 Packing Group II

**TDG**

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.  
 Hazard Class 3  
 Packing Group II

**IATA**

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.\*  
 Hazard Class 3  
 Packing Group II

**IMDG/IMO**

UN-No UN3271  
 Proper Shipping Name ETHERS, N.O.S.  
 Hazard Class 3  
 Packing Group II

## 15. Regulatory information

**United States of America Inventory**

| Component                        | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------------|----------|------|-----------------------------------------------|-----------------------------|
| Ethene, (2,2,2-trifluoroethoxy)- | 406-90-6 | X    | ACTIVE                                        | -                           |

**Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b) - Notices of Export**

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                        | CAS No   | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|----------------------------------|----------|-----|-----|-----------|-------|------|------|------|-------|------|
| Ethene, (2,2,2-trifluoroethoxy)- | 406-90-6 | -   | X   | 206-977-1 | -     | -    |      | -    | -     | -    |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                        | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|----------------------------------|---------------|------------|--------------|----------|--------------|
| Ethene, (2,2,2-trifluoroethoxy)- | -             | X          | -            | -        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component                        | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ethene, (2,2,2-trifluoroethoxy)- | 406-90-6 | -                                                                   | -                                                                             | -                                                                                                     |

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                        | CAS No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Ethene, (2,2,2-trifluoroethoxy)- | 406-90-6 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

See table for values

| Component                                             | OECD PFAS | US (EPA) PFAS | EU (ECHA) PFAS | UK (HSE) PFAS | Chemsec PFAS (Sin List) |
|-------------------------------------------------------|-----------|---------------|----------------|---------------|-------------------------|
| Ethene, (2,2,2-trifluoroethoxy)-<br>(CAS #: 406-90-6) | -         | -             | Listed         | Listed        | -                       |

**PFAS Legend**

Listed = Meets the PFAS definition of the named authority

**Other International Regulations**

| Component                           | CAS No   | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|-------------------------------------|----------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Ethene,<br>(2,2,2-trifluoroethoxy)- | 406-90-6 | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Not applicable                        |

**16. Other information****Prepared By**

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date**

30-Mar-2024

**Print Date**

30-Mar-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**